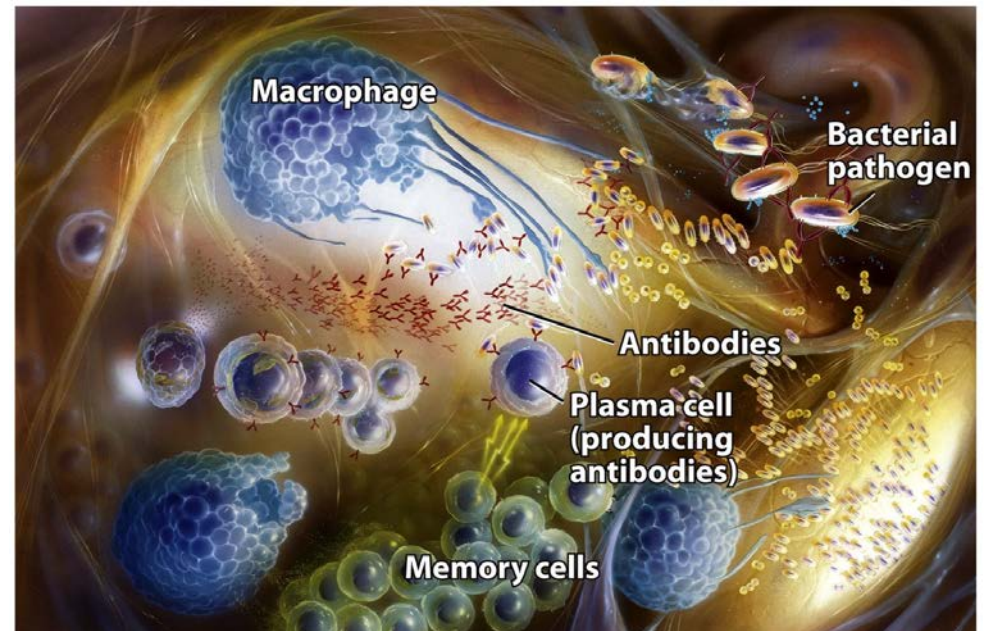
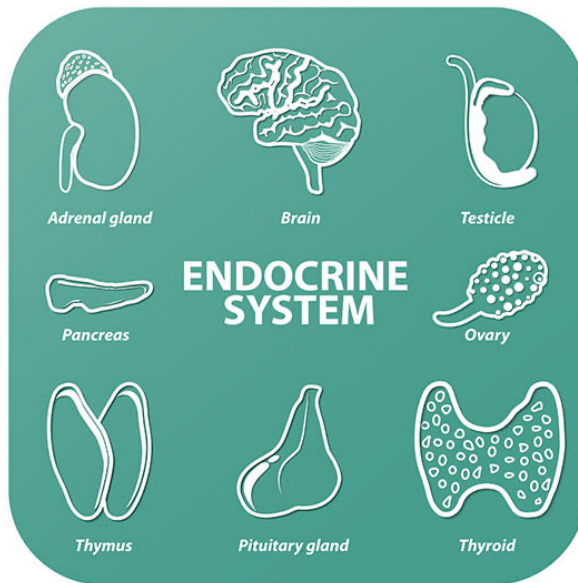


Endocrine System & Bodily Defenses

Instructor – Jeff Schinske



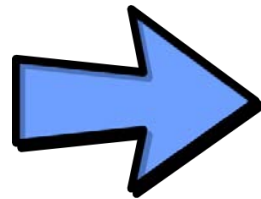


Bio 40C Proposed Norms

- Seek to enjoy your colleagues and your experiences in class!
- Actively select language that is inclusive, respectful, and welcoming
- Assume good intentions...no heroes and no villains (in this class)
- Recognize that we all have strengths as well as areas for growth
- Be curious, be skeptical
- Ask for help (and expect to receive it!)

Share with a partner...

1. Name
2. Something important to know about you as a person
3. What you remember from our studies of "mystery" hormones on Canvas



Person that traveled the farthest to get to Foothill today goes first



Debriefing Last Week

○ Last Week's Canvas Work

- A variety of different ideas about what the different "mystery" hormones might be.
- For some initial advice on revisions for recent application questions, check the grading keys posted underneath the prompts and the lecture recordings where we discussed the questions!



A Brief Preview of the Week

- Today
 - Reviewing Mystery Hormones & Endocrine Feedback Loops
 - Introducing the cells and chemicals of the body's defenses
- Lab Today
 - Endocrine System
- Online Lecture Work This Week
 - Endocrine System Case Studies!



What will you be able to do?

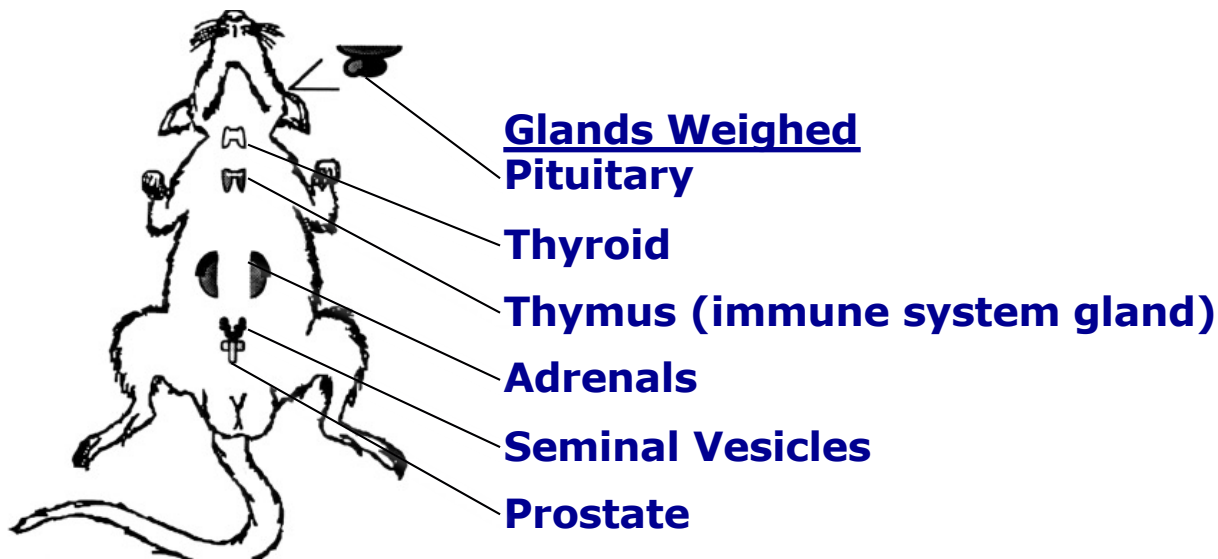
After class, you will be able to:

- Interpret experimental data to identify mystery hormones injected into the body.
- Demonstrate proficiency with critical thinking in the context of physiological feedback loops.
- Justify health recommendations regarding handwashing using information about the body's defenses.

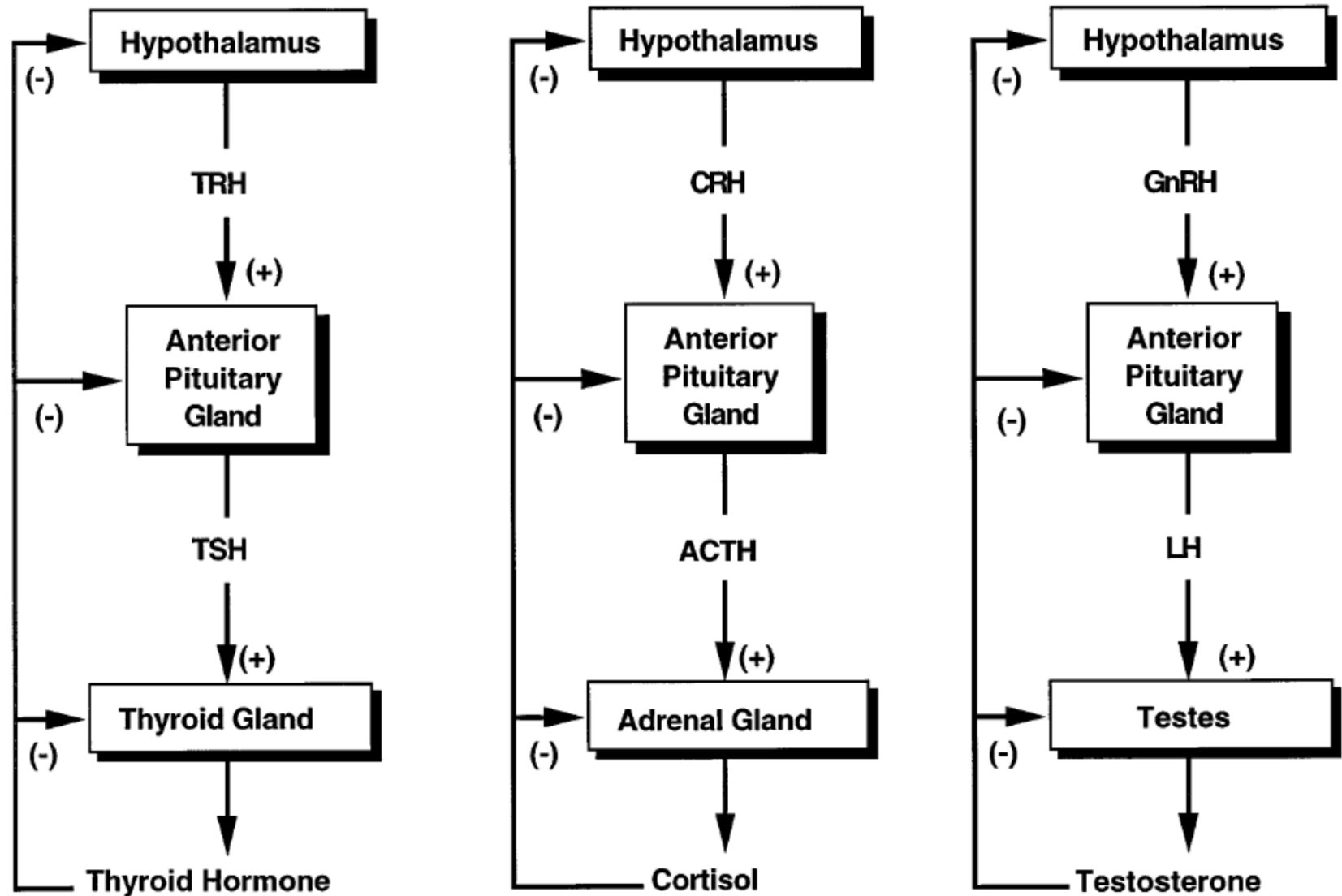
Reviewing Hormone Brainteasers!

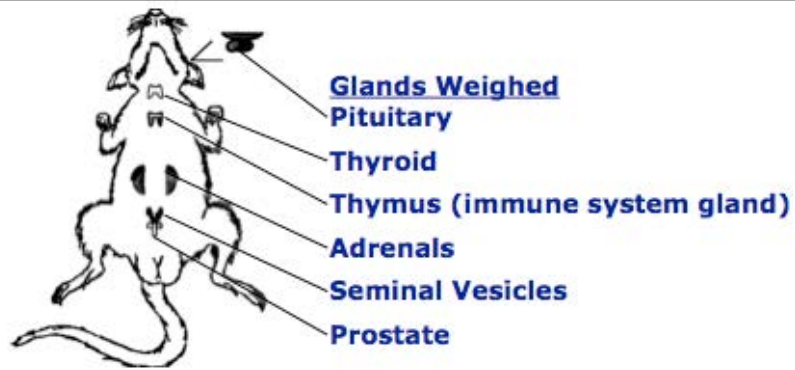
Antonia is a researcher studying the effects of hormones. She wants to see whether certain injected hormones will cause growth or shrinking in endocrine organs. Antonia injects groups of rats with either ACTH, Cortisol, TSH, or TRH. Antonia also leaves some rats alone, without injecting them with hormones (as a control).

Unfortunately, Antonia's notes go missing overnight and she no longer has a record for which hormones she puts in which rats! She measures the weights of all the rats' glands to see if she can figure out which hormones were put in which rats.



Reviewing Hormone Brainteasers!





***Example from
Module Discussion***

**Which of these was injected?
Support your choice by
discussing the components of
a hormone feedback loop.**

**ACTH
Cortisol
LH
TSH
TRH
Testosterone**



Control Intact

Pituitary: 12.9 mg

Thyroid: 250 mg

Thymus: 475 mg

Adrenals: 40 mg

Seminal vesicles: 500 mg

Prostate: 425 mg

Testes: 3200 mg

Body weight: 300 g

Injected Rat B

Pituitary: 10.1 mg

Thyroid: 245 mg

Thymus: 250 mg

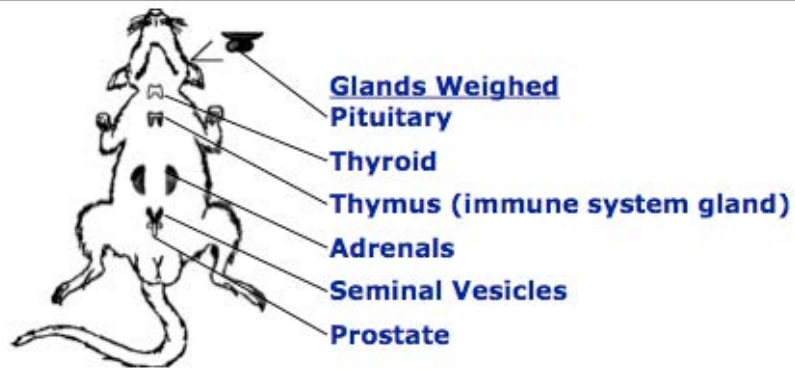
Adrenals: 100 mg

Seminal vesicles: 490 mg

Prostate: 430 mg

Testes: 3000 mg

Body weight: 200 g



New Example

Which of these was injected?
Support your choice by
discussing the components of
a hormone feedback loop.

ACTH
Cortisol
LH
TSH
TRH
Testosterone



Control Intact

Pituitary: 12.9 mg

Thyroid: 250 mg

Thymus: 475 mg

Adrenals: 40 mg

Seminal vesicles: 500 mg

Prostate: 425 mg

Testes: 3200 mg

Body weight: 300 g

Injected Rat F

Pituitary: 10.2 mg

Thyroid: 252 mg

Thymus: 470 mg

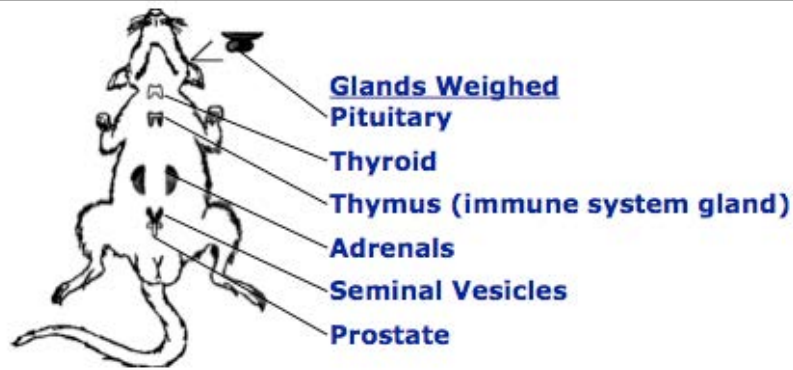
Adrenals: 38 mg

Seminal vesicles: 1400 mg

Prostate: 900 mg

Testes: 2400 mg

Body weight: 490 g



***Trying a new one
on your own!***

**Which of these was injected?
Support your choice by
discussing the components of
a hormone feedback loop.**

**ACTH
Cortisol
LH
TSH
TRH
Testosterone**



Control Intact

Pituitary: 12.9 mg

Thyroid: 250 mg

Thymus: 475 mg

Adrenals: 40 mg

Seminal vesicles: 500 mg

Prostate: 425 mg

Testes: 3200 mg

Body weight: 300 g

Injected Rat

Pituitary: 8 mg

Thyroid: 500 mg

Thymus: 455 mg

Adrenals: 37 mg

Seminal vesicles: 480 mg

Prostate: 405 mg

Testes: 2790 mg

Body weight: 152 g

11 **Your Name**

Your responses...



What will you be able to do?

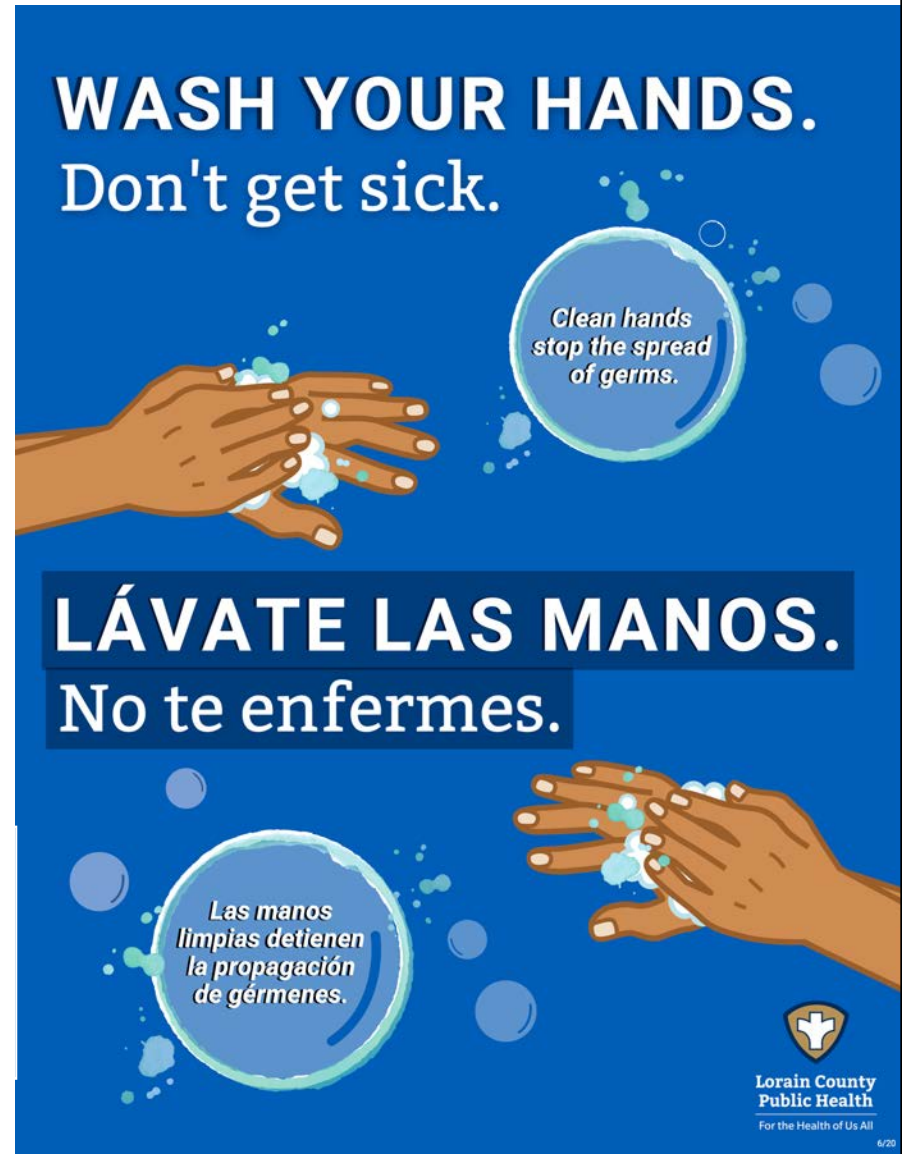
After class, you will be able to:

- Interpret experimental data to identify mystery hormones injected into the body.
- Demonstrate proficiency with critical thinking in the context of physiological feedback loops.
- Justify health recommendations regarding handwashing using information about the body's defenses.

Why do we so often see these recommendations?
What would happen if you do touch your face and transfer germs there?

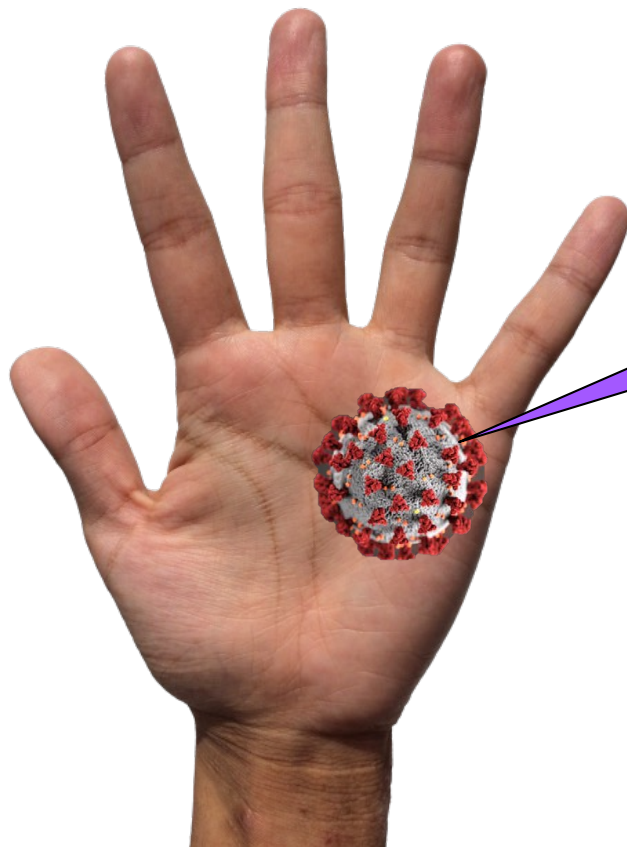


Remember, in A&P we are aiming to address this from the cellular and chemical perspectives!



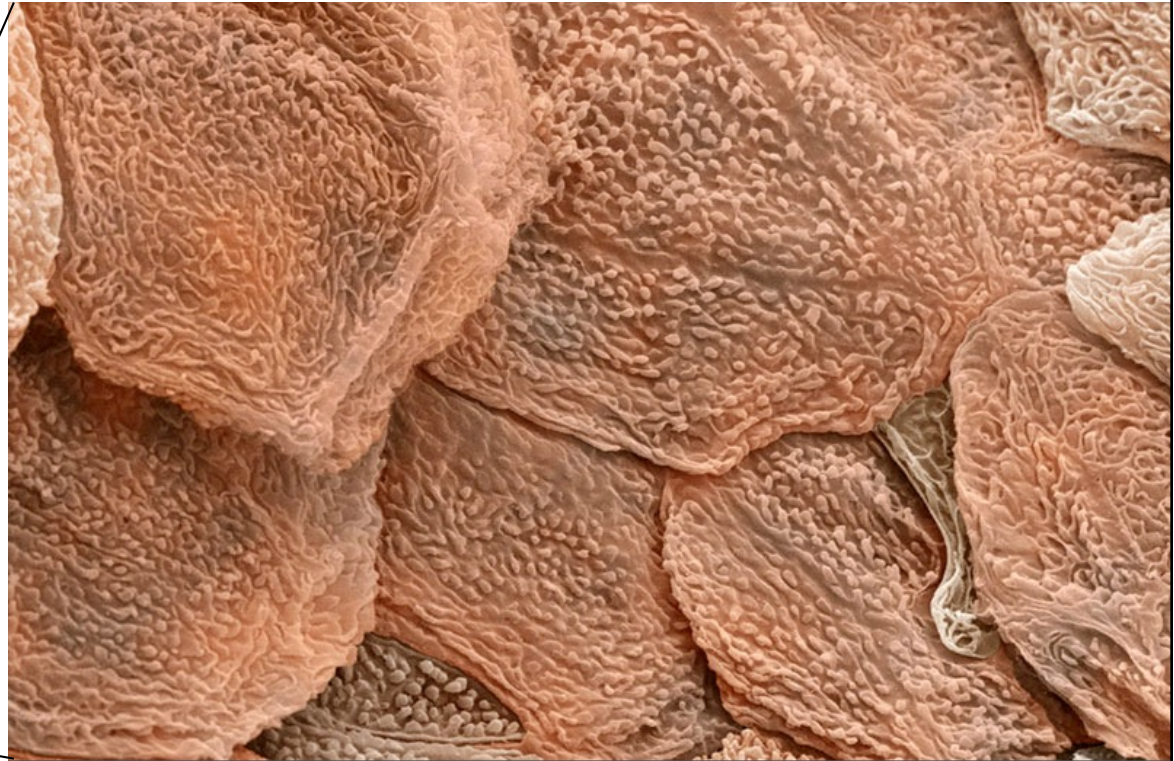
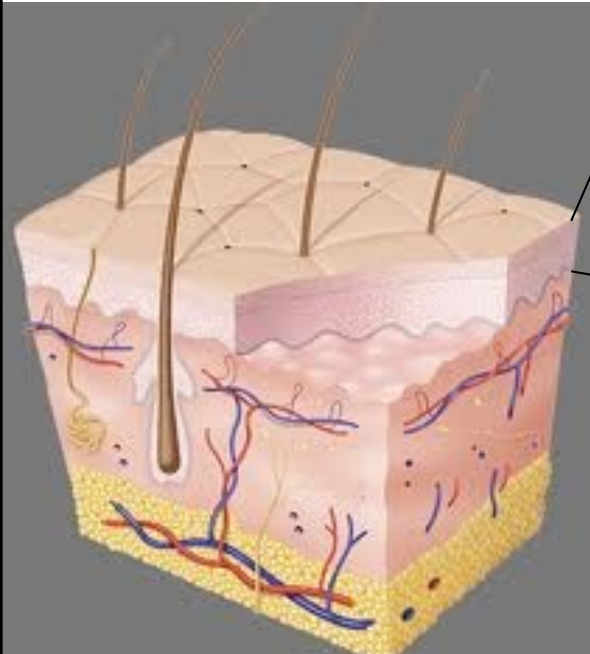
Experiencing our nonspecific/innate defenses from the perspective of a virus!

We're going to adopt the perspective of a virus particle resting on the surface of your hand and see what the particle would experience! In this way, we will achieve some of our goals of understanding the cellular and molecular aspects of your defenses.



Lovely to meet you! I'm a virus that can cause you to get a cold!

Keratinized Epithelium: Skin

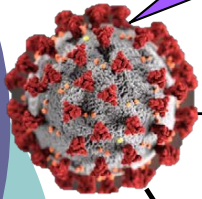


Scanning electron micrograph of epidermis
Photograph by Andrew Syred/Science Photo Library

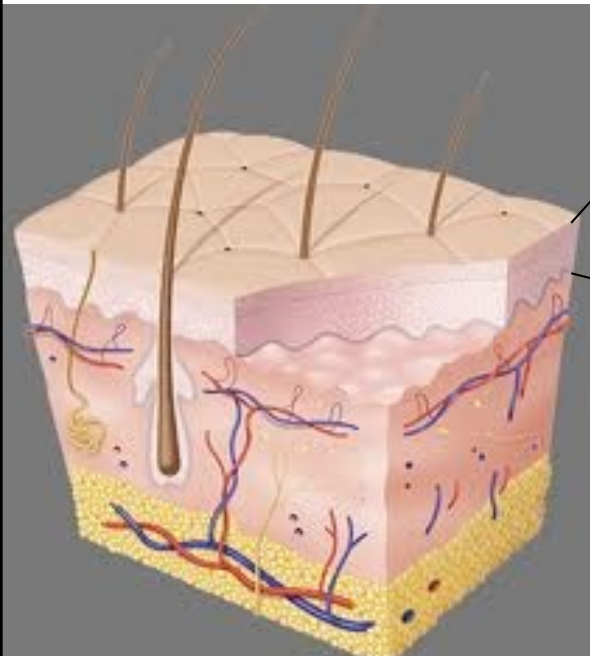
NATIONAL
GEOGRAPHIC

Keratin is Durable & Cells Have Tight Junctions (Adherens Junctions, Desmosomes)

Hey! Let me in!

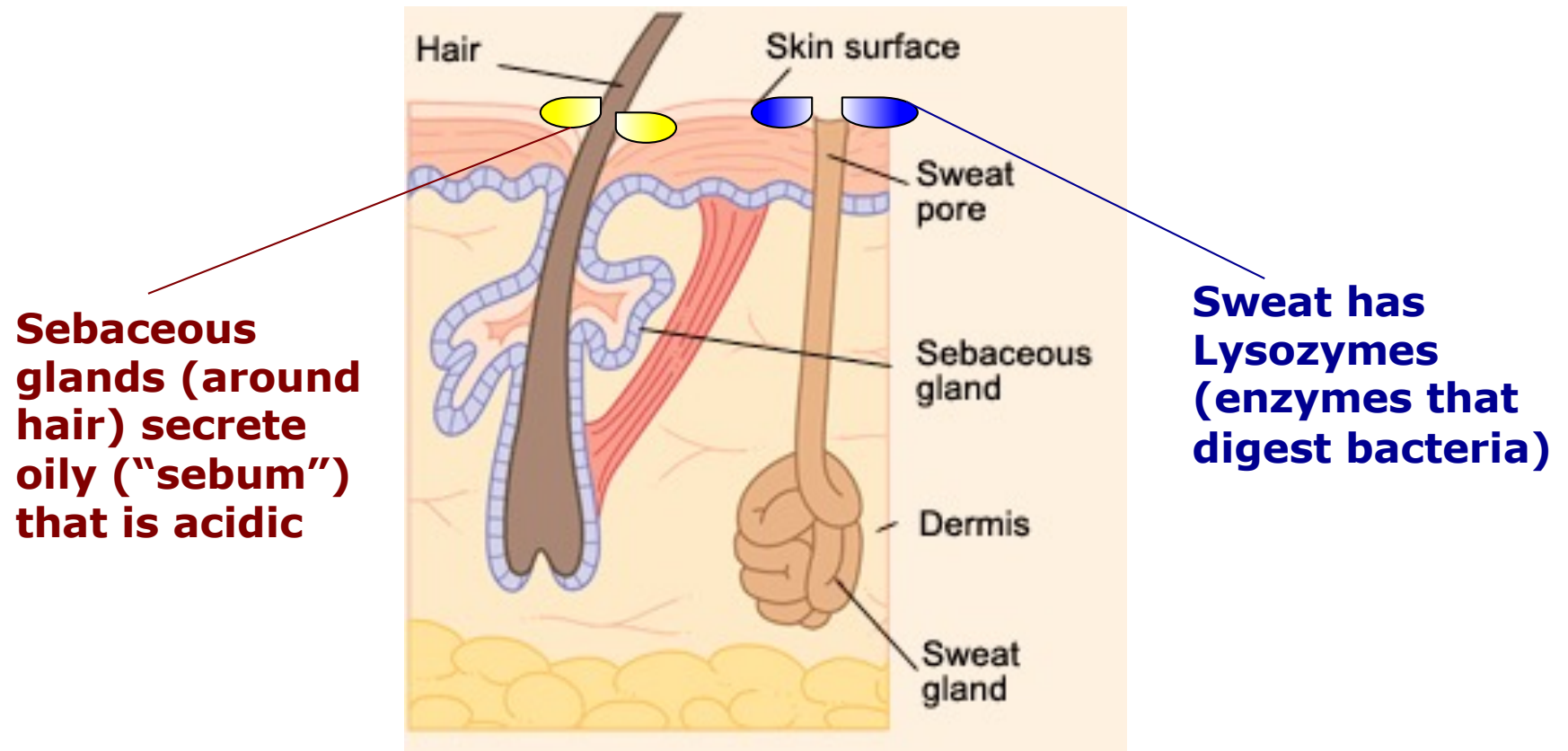


“As long as you wash your hands before touching your face, you should be OK, because viral droplets don’t pass through skin.”



Keratin is Durable & Cells Have Tight Junctions (Adherens Junctions, Desmosomes)

Sebum & Sweat Are Antimicrobial

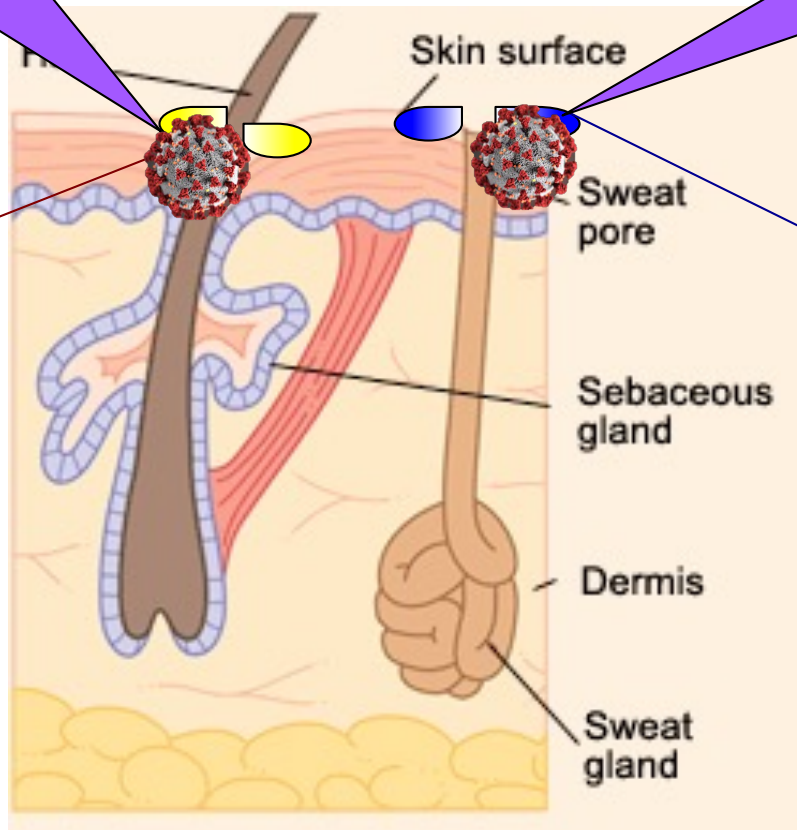


Sebum & Sweat Are Antimicrobial

Ack! I do best in a pH of ~ 6 , but this can be as low as 4.5!

I'm not a bacterium, so no worries here!

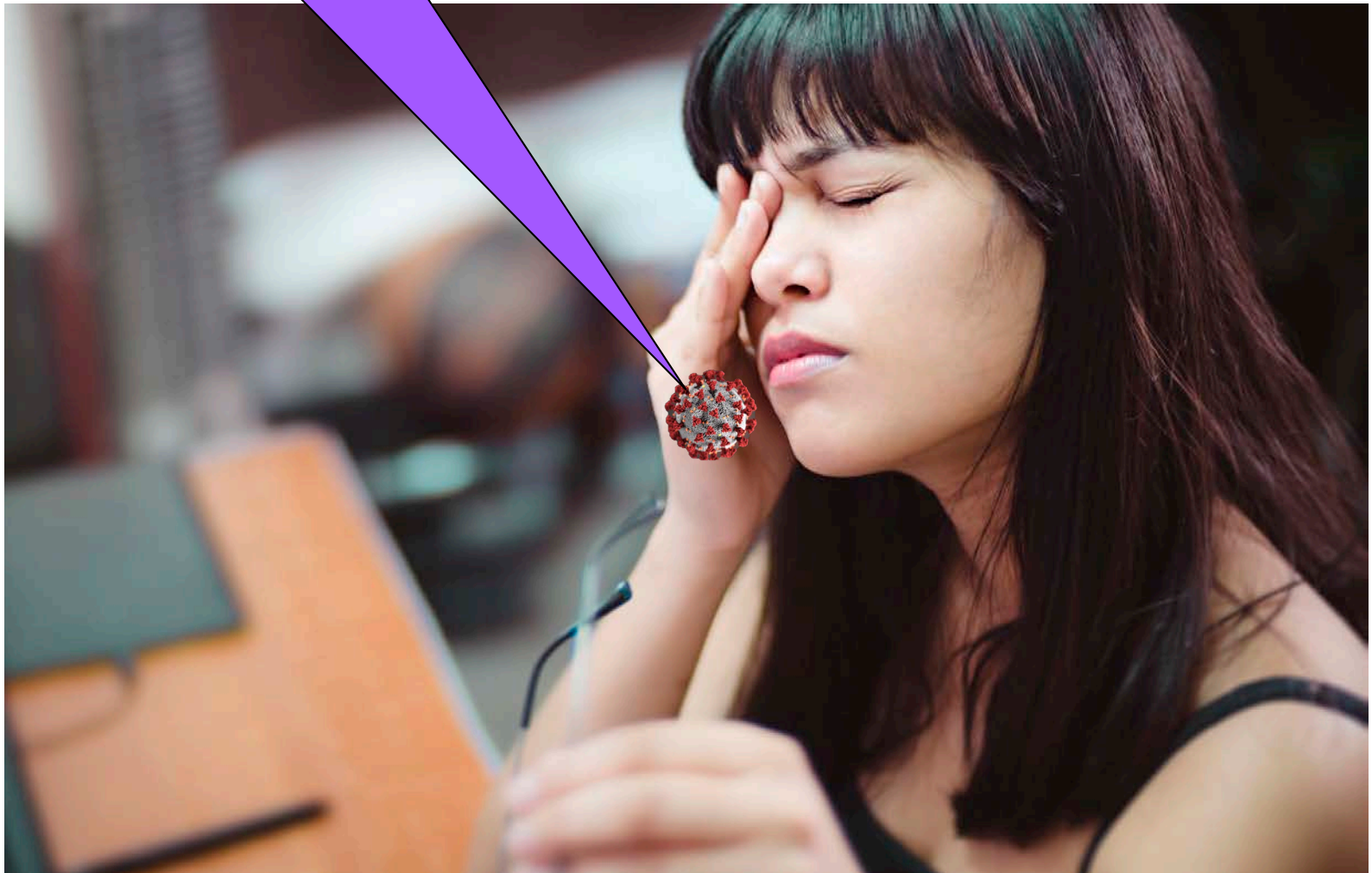
Sebaceous glands (around hair) secrete oily ("sebum") that is acidic



Sweat has Lysozymes (enzymes that digest bacteria)

So the skin is a quite formidable defense! But...

**Yes! That's right!
Touch the face!!!**



Sticky Mucus Lines Many Passages & Cilia Push the Mucus



Hair-like projections called cilia line the primary bronchus to remove microbes and debris from the interior of the lungs

Cilia

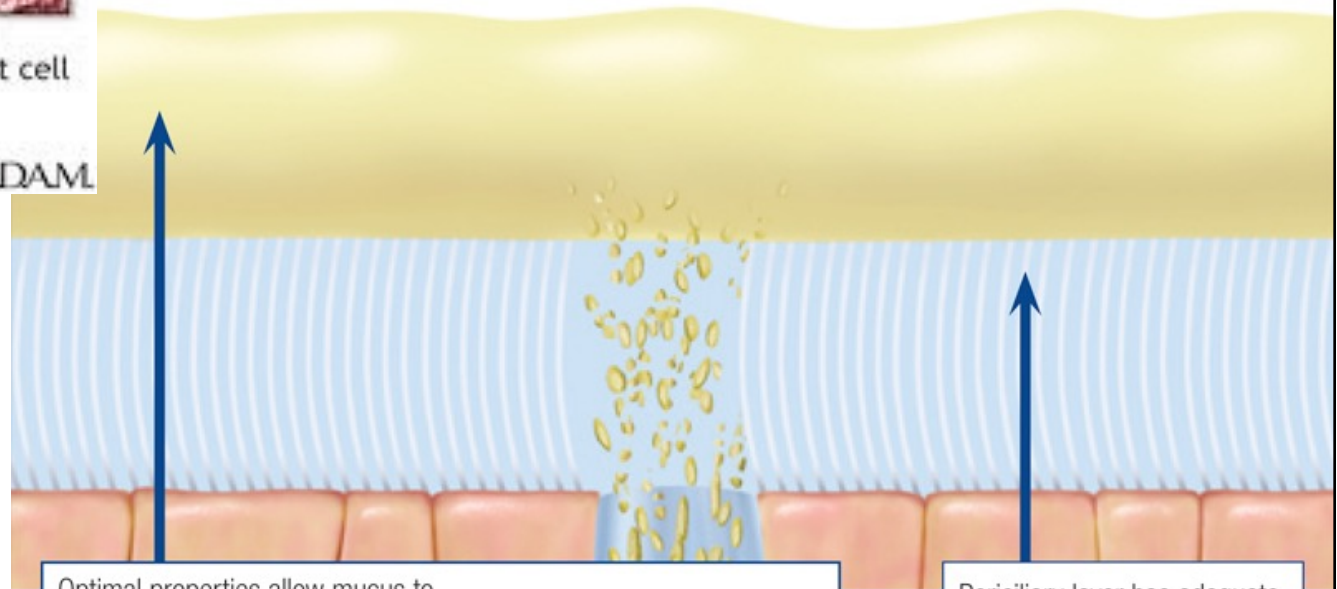
Primary bronchus



Goblet cell

ADAM

Normal mucus can be easily transported¹⁻⁵



Optimal properties allow mucus to

- Trap inhaled pathogens, allergens, and irritants
- Be easily transported out of the lungs via cough and ciliary beating

Periciliary layer has adequate volume to allow rhythmic ciliary beating

Sticky Mucus Lines Many Passages & Cilia Push the Mucus



Hair-like projections called cilia line the primary bronchus to remove microbes and debris from the interior of the lungs

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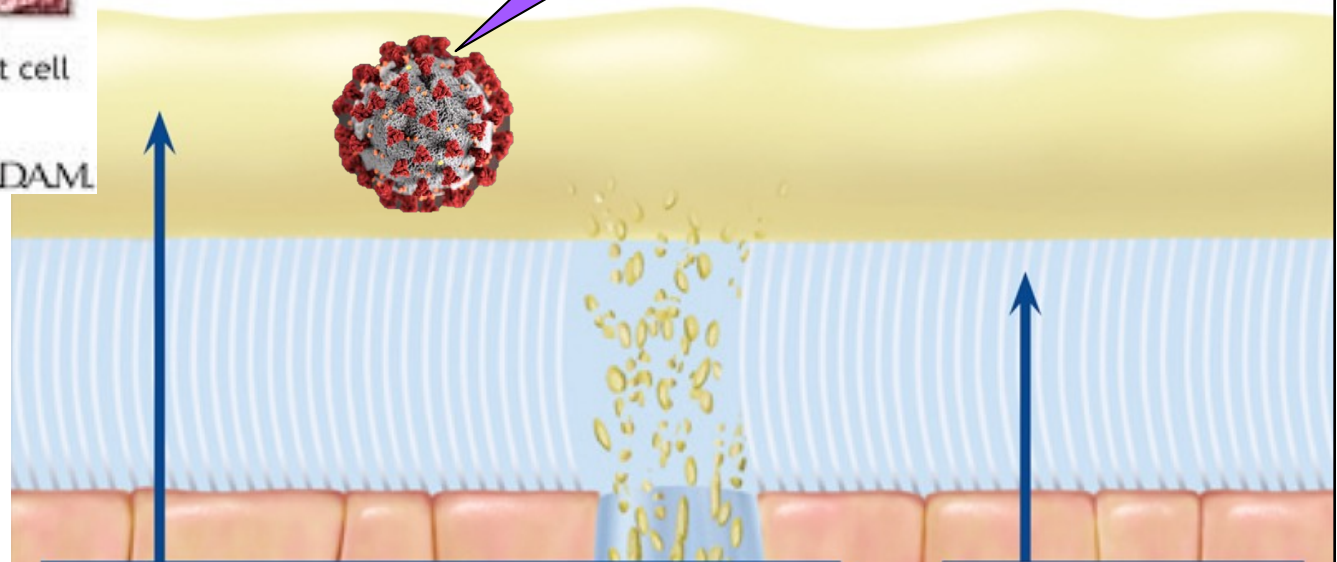


Goblet cell

ADAM

**Well this is just gross!
And where are you
pushing me?**

Normal mucus can be easily transported¹⁻⁵



Optimal properties allow mucus to

- Trap inhaled pathogens, allergens, and irritants
- Be easily transported out of the lungs via cough and ciliary beating

Periciliary layer has adequate volume to allow rhythmic ciliary beating

Coughing or Sneezing Ejects Pathogens



Coughing or Sneezing Ejects Pathogens



Sticky Mucus Lines Many Passages & Cilia Push the Mucus



Hair-like projections called cilia line the primary bronchus to remove microbes and debris from the interior of the lungs

Cilia

Primary bronchus

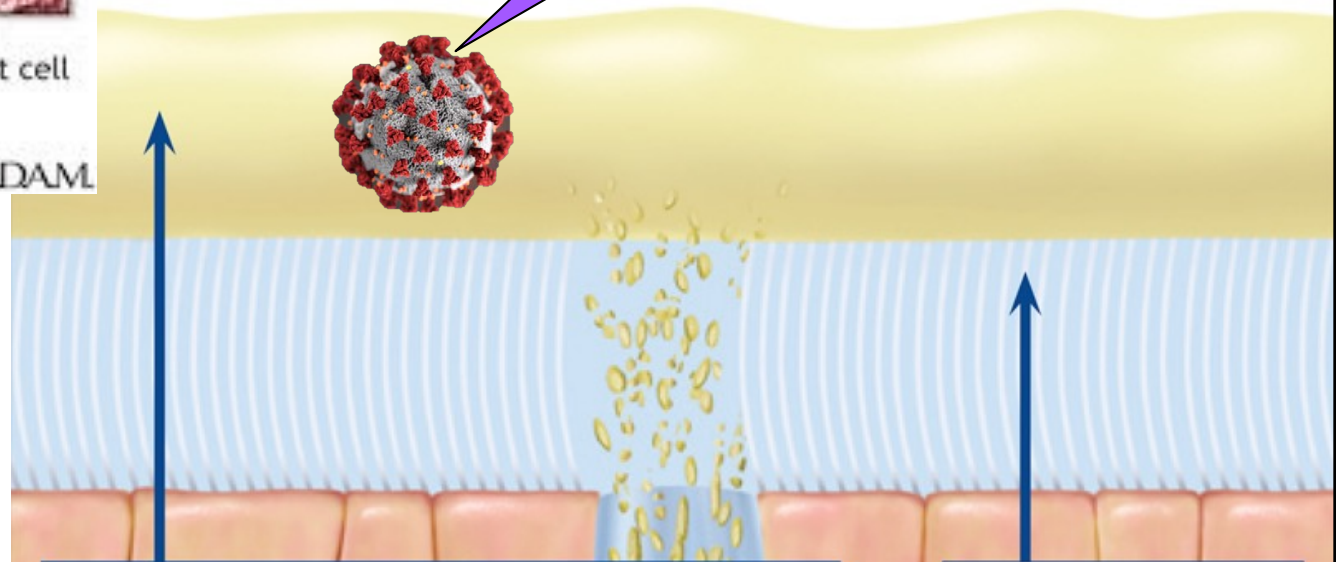


Goblet cell

ADAM

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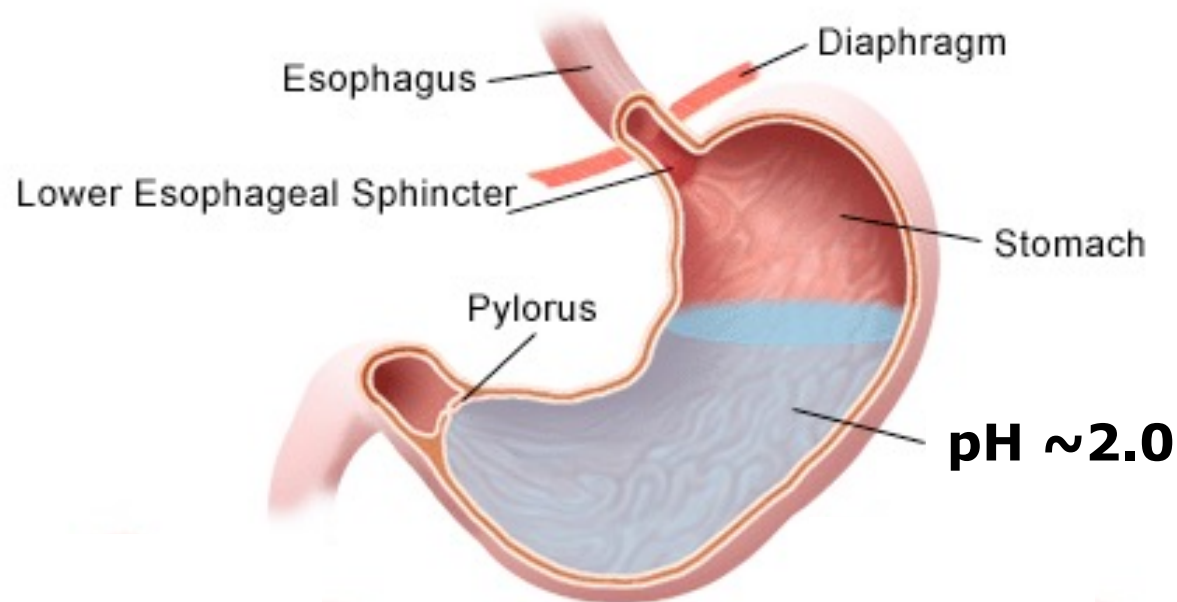


Optimal properties allow mucus to

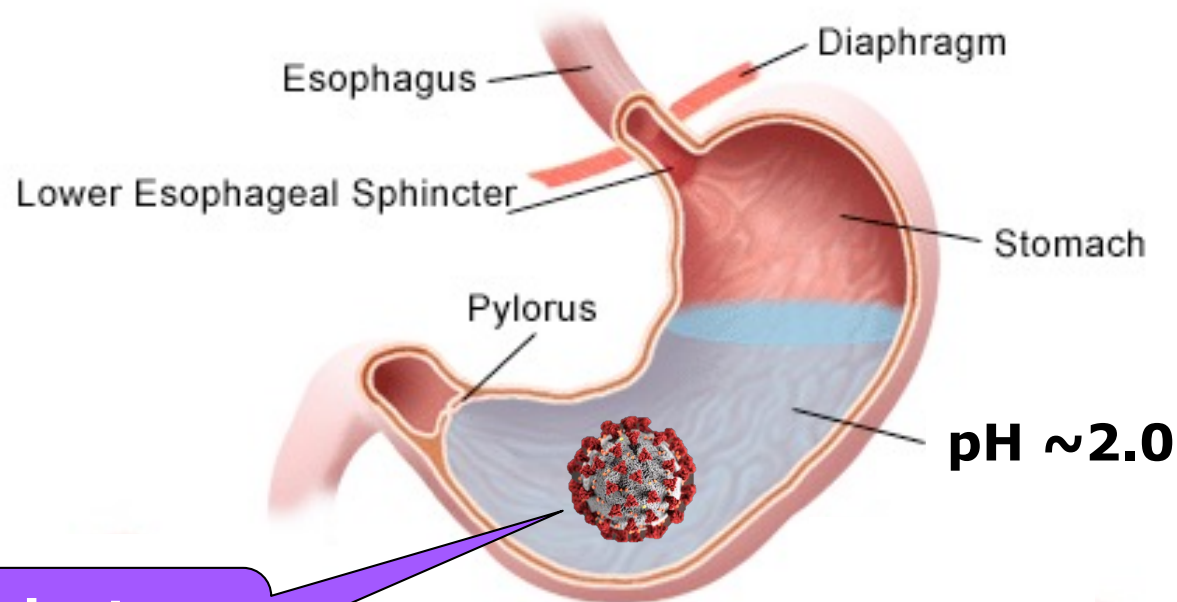
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Periciliary layer has adequate volume to allow rhythmic ciliary beating

Acidic Gastric Juices Protect One of the Main Body Entrances

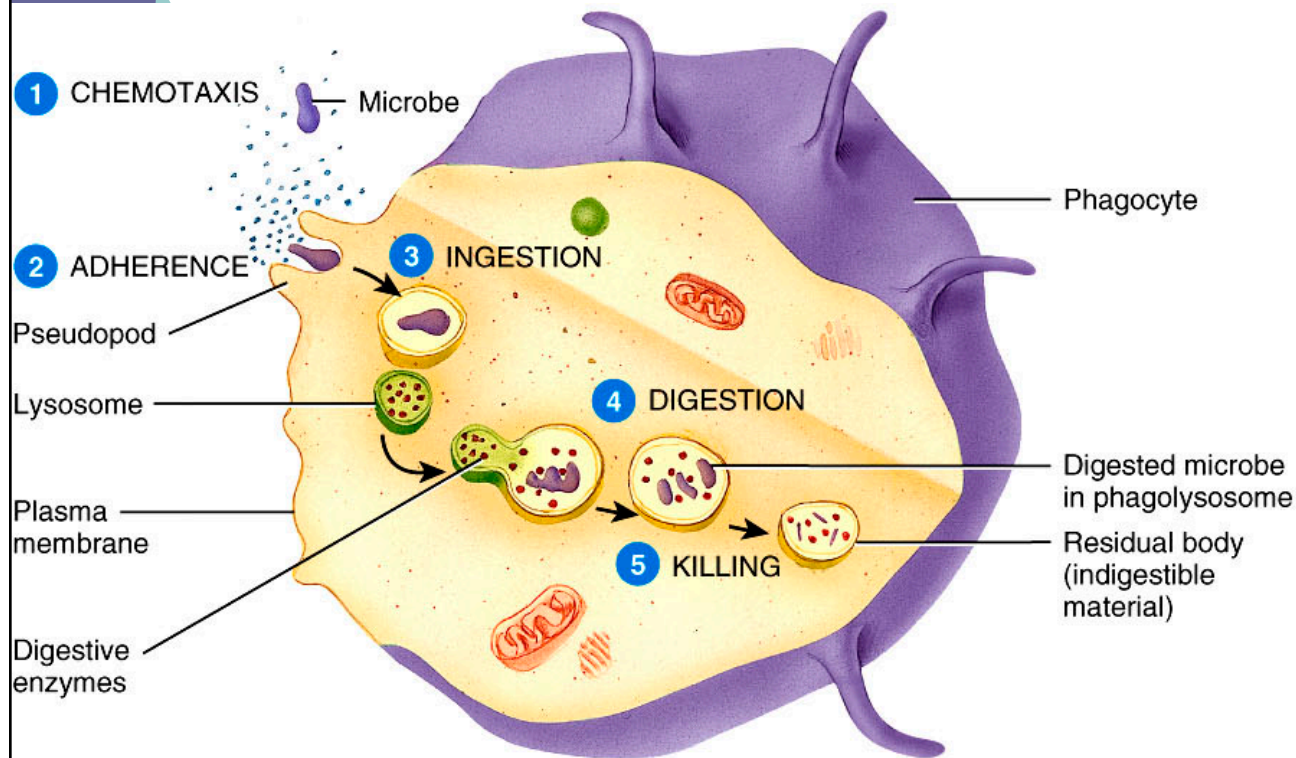


Acidic Gastric Juices Protect One of the Main Body Entrances

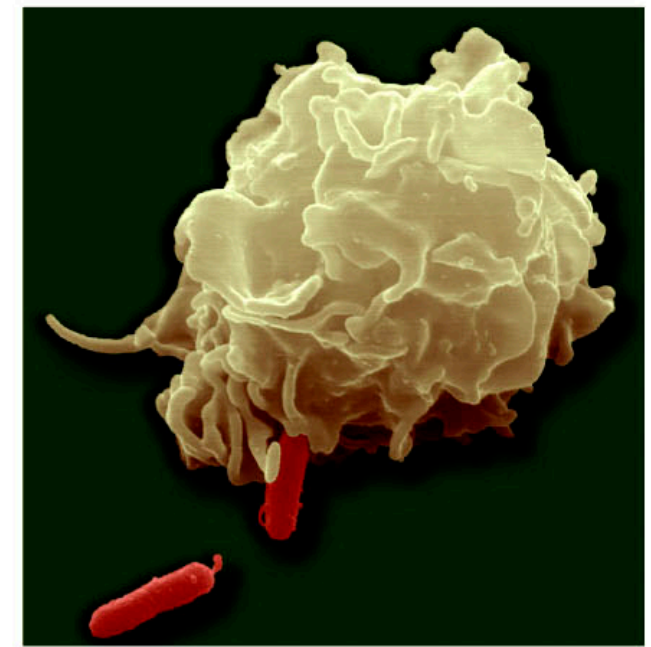


**Now this pH is just
way too low!!!**

Phagocytes Ingest (“Eat”) Pathogens



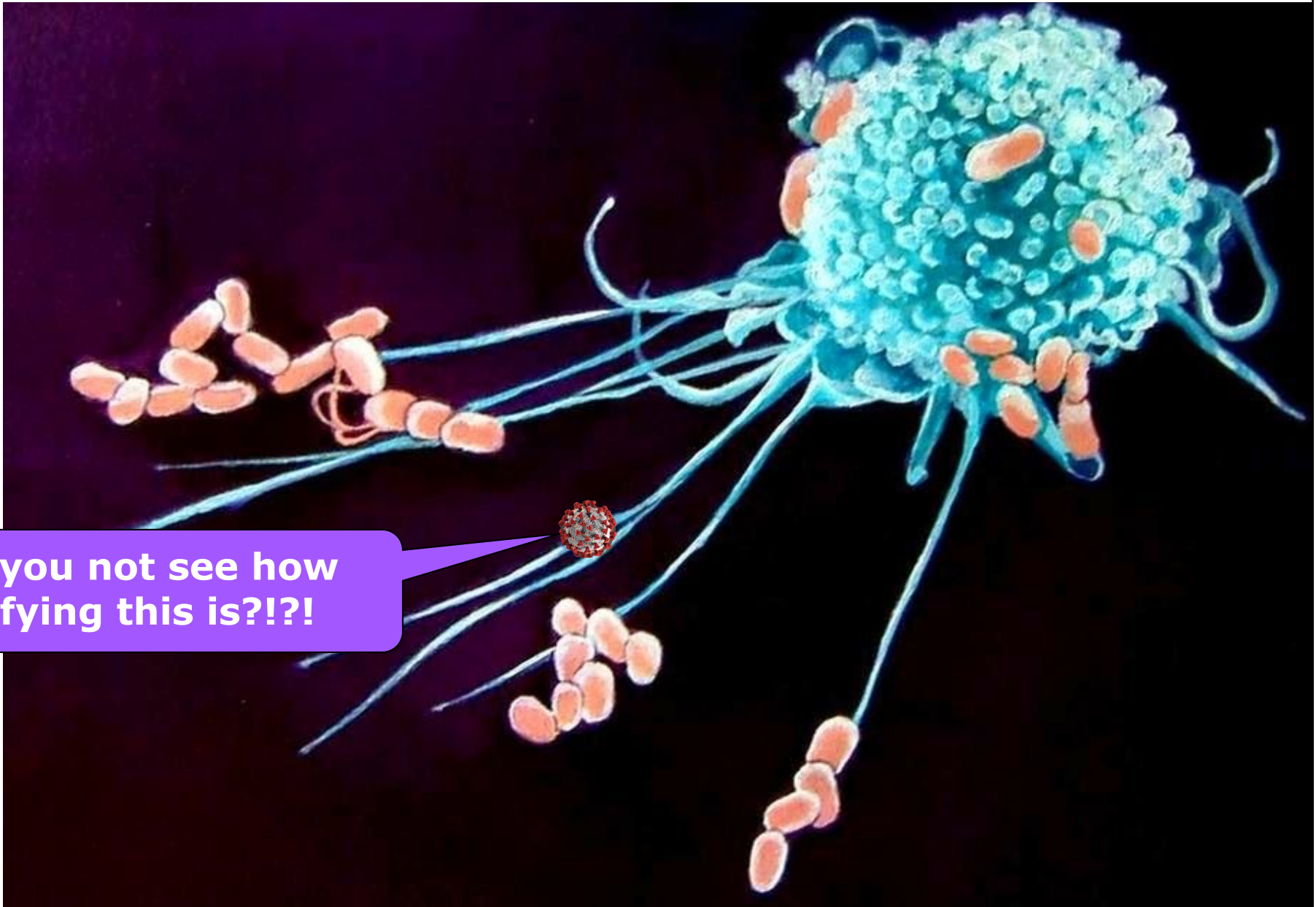
(a) Phases of phagocytosis



(b) Phagocyte (white blood cell) engulfing a microbe.

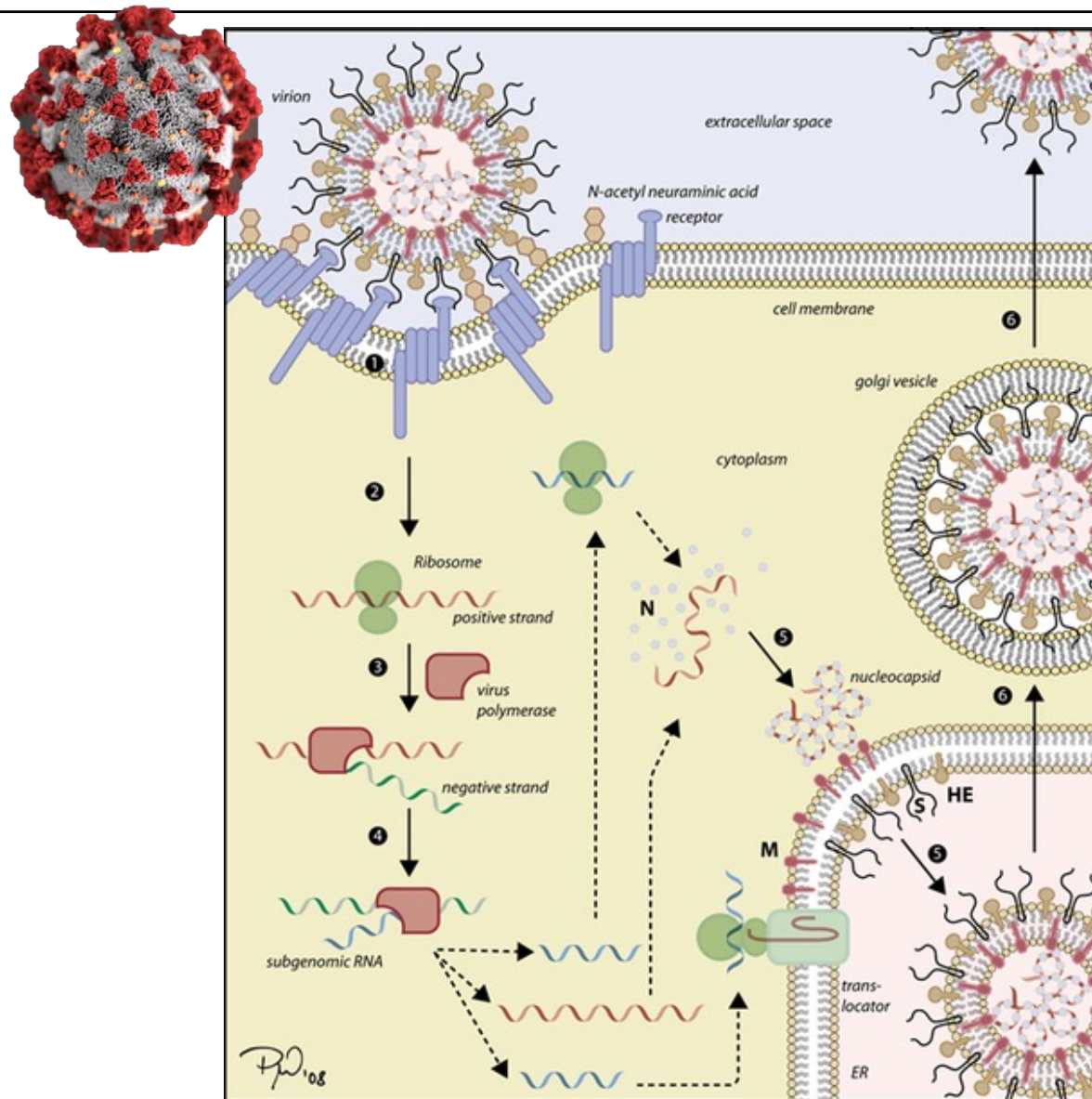
<https://www.youtube.com/watch?v=ygkvXT1BmaA>

Phagocytes Ingest (“Eat”) Pathogens

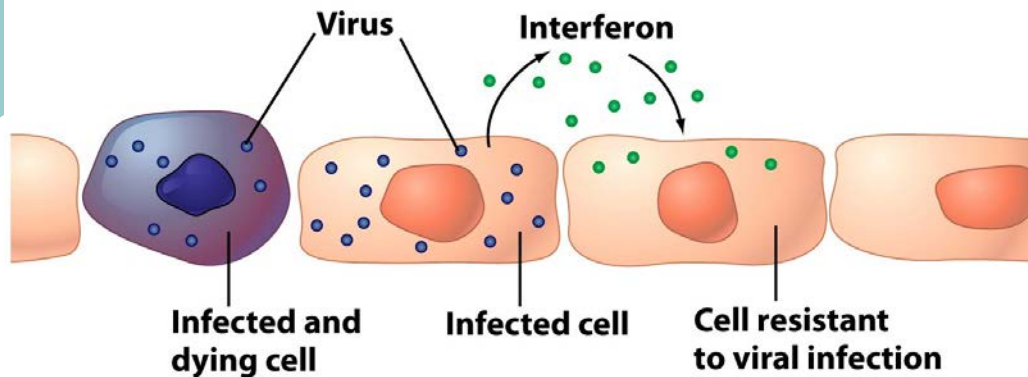


Can you not see how terrifying this is?!?!

Infection of a respiratory cell



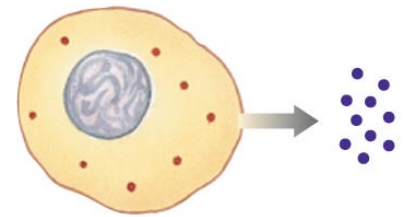
Interferons Send Messages About Viral Infection to Neighboring Cells & Slow Inflammation



Three of the types of interferons

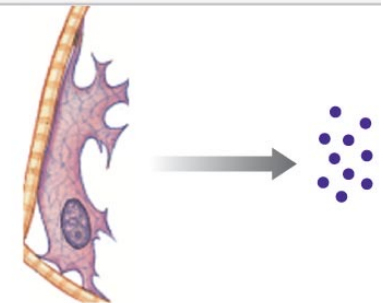
1

Alpha (α)-interferons are produced by cells infected with viruses. They attract and stimulate NK cells and enhance resistance to viral infection.



2

Beta (β)-interferons, secreted by fibroblasts, slow inflammation in a damaged area.

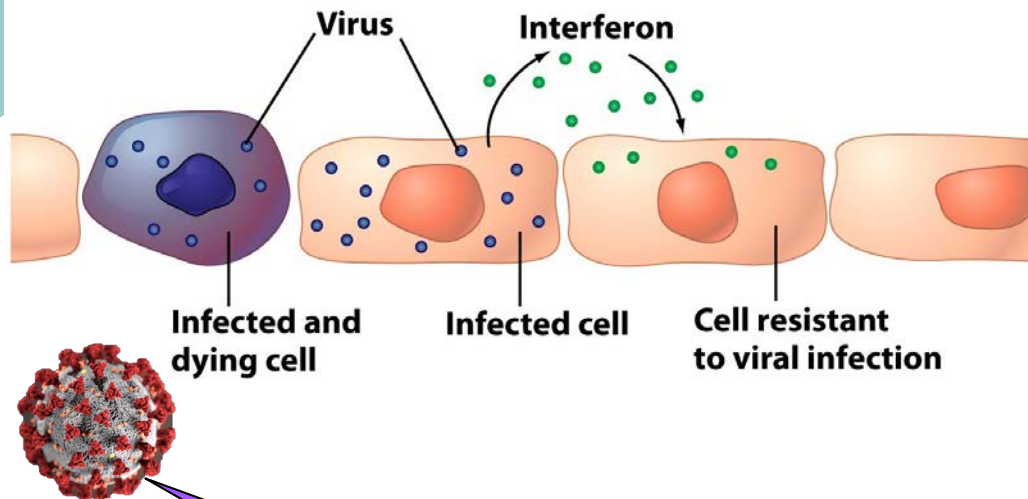


3

Gamma (γ)-interferons, secreted by T cells and NK cells, stimulate macrophage activity.



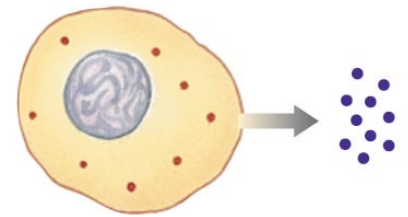
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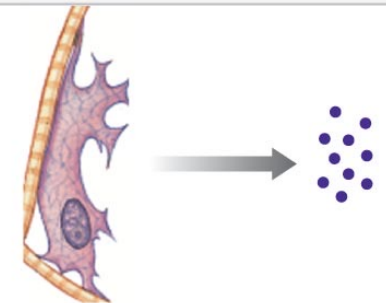
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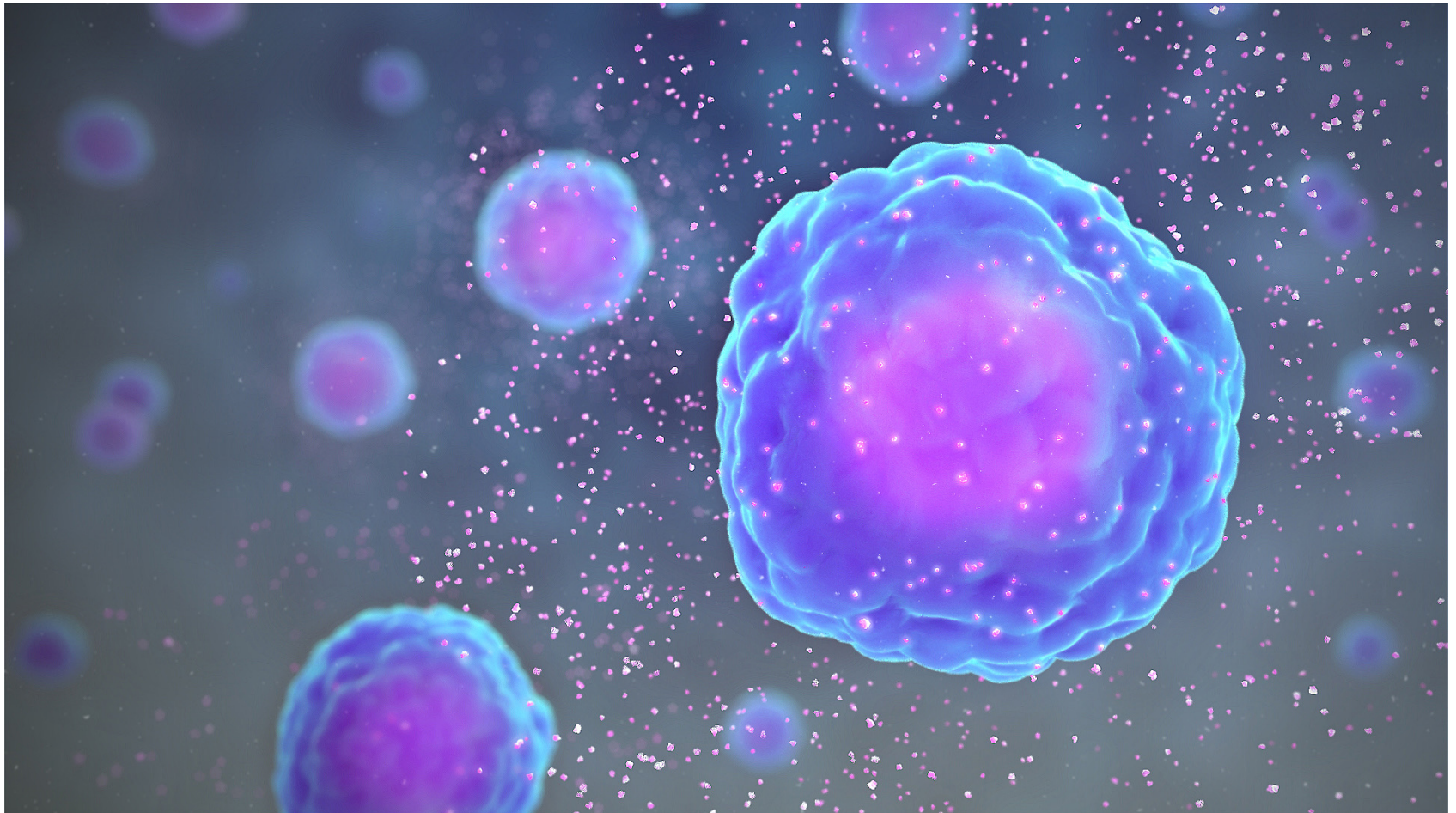


3

Gamma (γ)-interferons, secreted by T cells and NK cells, stimulate macrophage activity.



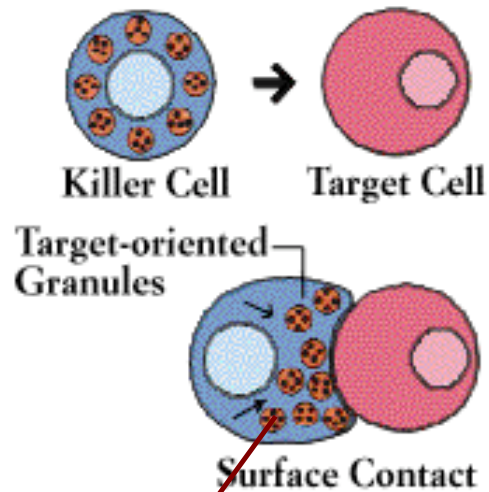
Interferons are one of many types of cytokines



“Cytokines are soluble proteins that are produced in response to a perceived foreign particle and function as chemical messengers for regulating the immune system.”

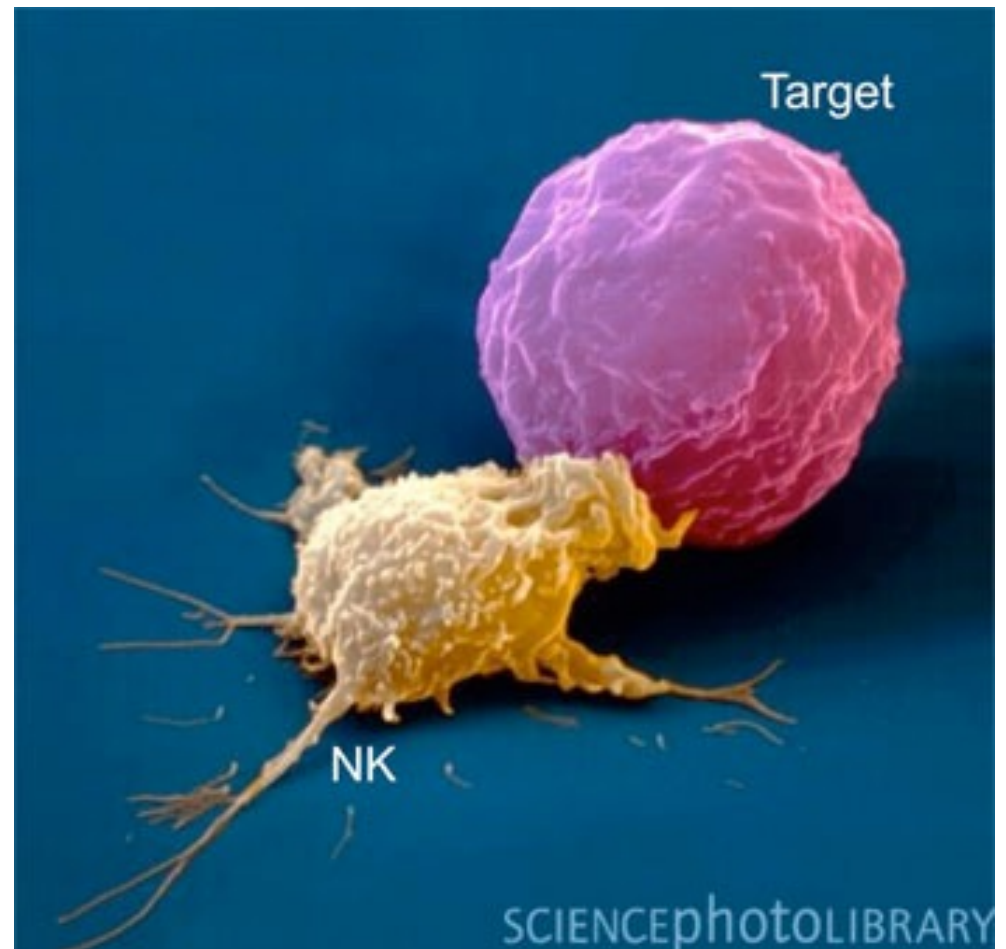
[https://bio.libretexts.org/Bookshelves/Microbiology/Book%3A_Microbiology_\(Kaiser\)/Unit_5%3A_Innate_Immunity/11.4%3A_Early_Induced_Innate_Immunity/11.3C%3A_Cytokines_Important_in_Innate_Immunity](https://bio.libretexts.org/Bookshelves/Microbiology/Book%3A_Microbiology_(Kaiser)/Unit_5%3A_Innate_Immunity/11.4%3A_Early_Induced_Innate_Immunity/11.3C%3A_Cytokines_Important_in_Innate_Immunity)

Natural Killer (NK) Cells Burst or Initiate Suicide in Abnormal (infected, cancerous) Body Cells

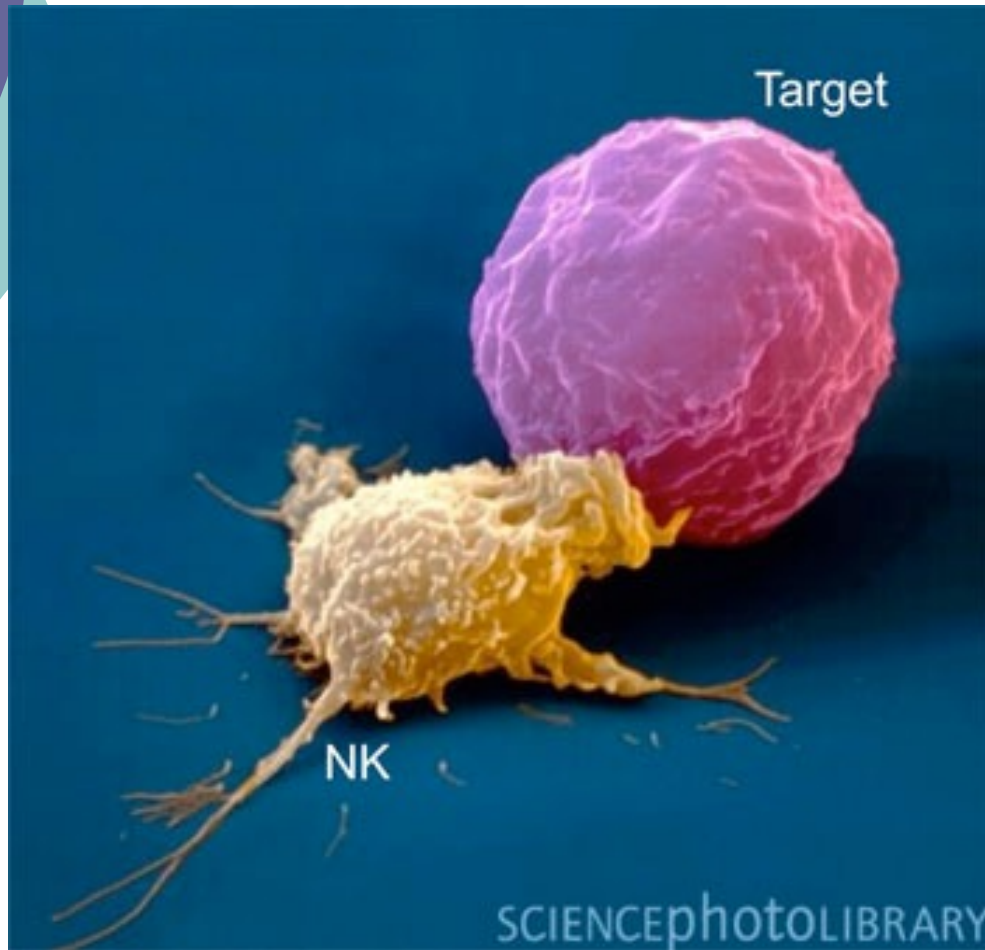


Granules contain chemicals to burst cell or cause it to "commit suicide" (apoptosis)

NK cell stays alive to look for other abnormal cells



A couple views of NK Cell Activity



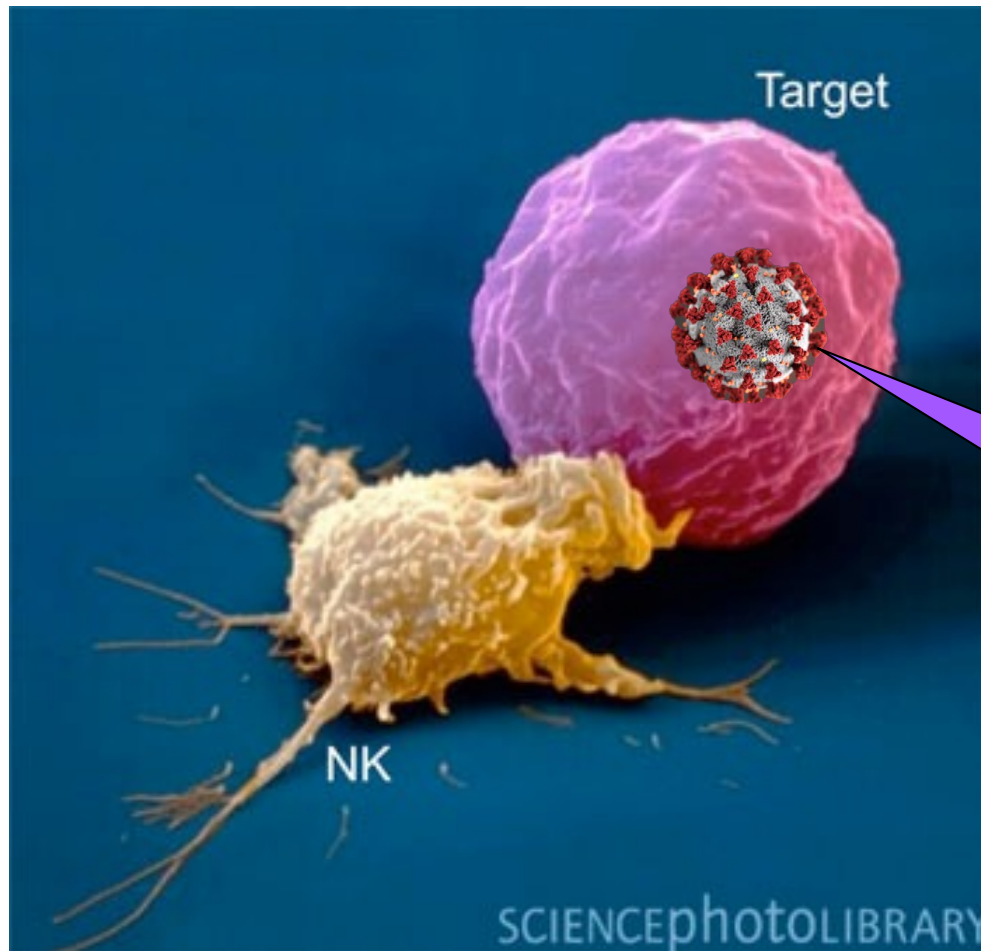
NK Cell Attack

<https://www.youtube.com/watch?v=HNP1EAYLhOs>

Or Perhaps...

<https://www.youtube.com/watch?v=fioU3uKYsdU>

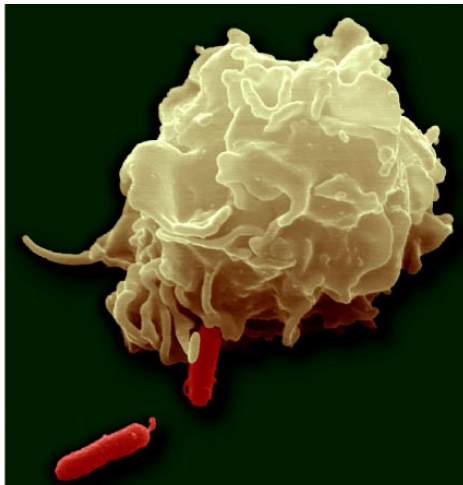
Natural Killer (NK) Cells Burst or Initiate Suicide in Abnormal (infected, cancerous) Body Cells



Hey! Back away from my host cell!

Thinking about the cells of “innate” immunity...

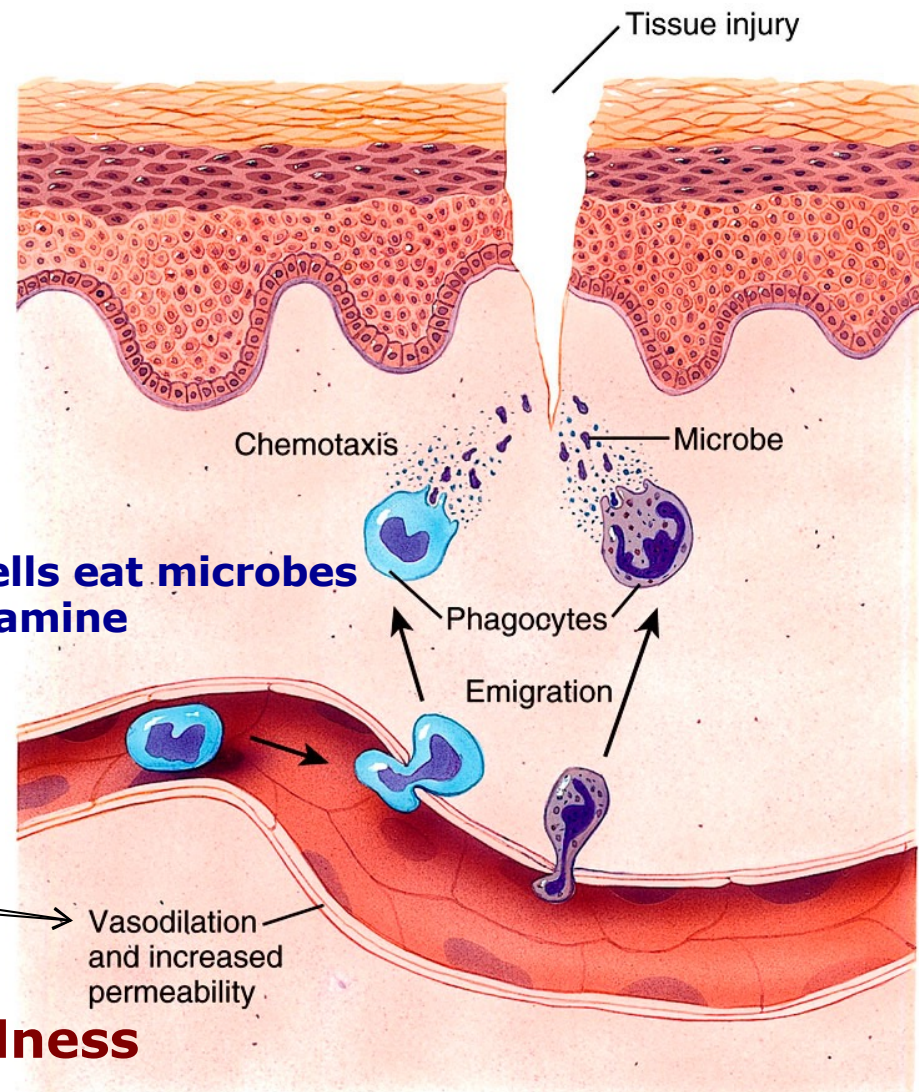
Compare and contrast the roles of **phagocytes** and **natural killer (NK) cells** in our immune defenses. How are the activities of those cells similar? How are they different?



Inflammation is a reaction to bring defenses & repairs to a specific area

Basophils & Mast Cells Produce
Histamine
Leukotrienes
Makes vessels leaky to
bring nutrients & more
defenses to area = swelling, redness

**Responding cells eat microbes
& release histamine**



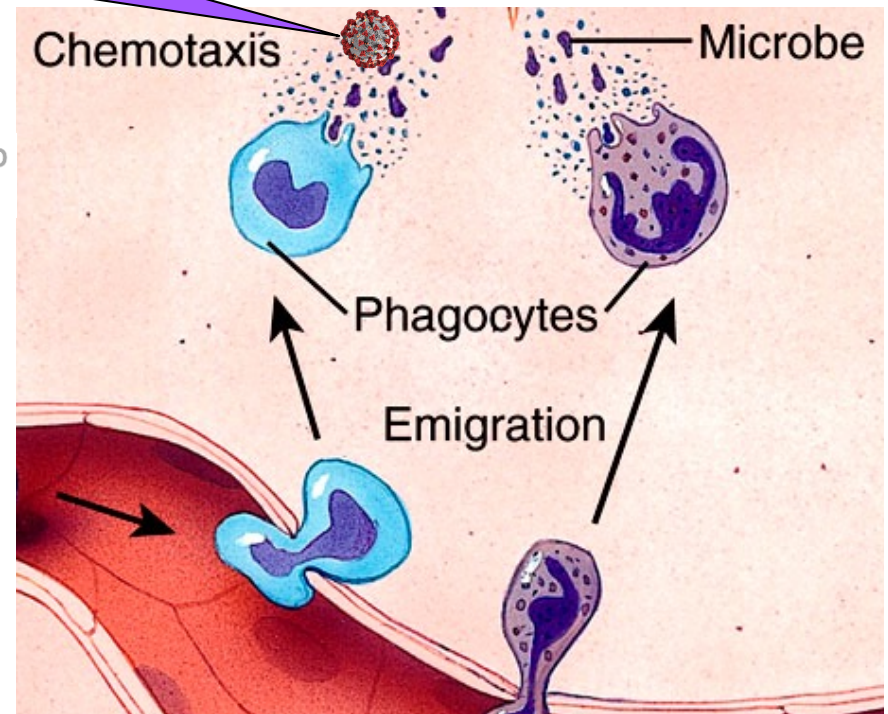
Phagocytes migrate from blood to site of tissue injury

Inflammation is a reaction to bring defenses & repairs to a specific area

Whoa! Why is everyone coming here?!

JBRHA 34, NO. 1, JANUARY-FEBRUARY, 2020 / JOURNAL OF BIOLOGICAL REGULATORS AND HOMEOSTATIC AGENTS

“Viruses are primarily attacked by immune cells including mast cells (MCs), which are located in the submucosa of the respiratory tract and in the nasal cavity and represent a barrier of protection against microorganisms. Viral activate MCs release early inflammatory chemical compounds including histamine and protease”



Fever is a whole-body reaction to speed up repairs

Possible Benefits:

- **White blood cells move faster**
- **White blood cells eat faster**
- **White blood cells divide faster**
- **Cytokines enhanced**



Fever is a whole-body reaction to speed up repairs

Possible Benefits:

- **White blood cells move faster**
- **White blood cells eat faster**
- **White blood cells divide faster**
- **Cytokines enhanced**

Uh oh! Why so many of you and what's the rush?!

